Individual Research Log Book

Decentralized Multi-Robot Collaboration: Scaling Vision-Language-Action Models from Single-Agent Manipulation to Communication-Free Teamwork

Course	CMPE 295A
Student	Kanaka Sarat Siripurapu
Email	kanakasarat.siripurapu@sjsu.edu
Advisor	Dr. Wencen Wu
Lane	Evaluation and data
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Why this log exists

I am on the evaluation and data side of this project. That means the dataset notes, the success metric, the evaluation wrapper, and the scored baselines. I did not set up Isaac Lab, I did not build the SmolVLA policy server, and I did not write the reference stack. Those are Ritwik, Snehil, and Bharat.

A lot of my work this month was waiting on them and then writing down exactly what I needed before I could score anything. That sounds passive. It was the actual bottleneck. If the cameras, action order, instruction string, or LeRobot version drift between runs, the success rate is junk no matter how clean the training looks.

Snapshot as of Oct 28

Finished on my lane	Run-metadata sheet, dataset audit, metric draft, wrapper skeleton + CSV schema, B0/B1 command plans, failure-label categories, Workbook 2 Ch 6/8/9 draft on feature/kanaka_workbook2.
Still blocked / waiting	Live B0 and B1 scores (need A40 rollout), final LeRobot pin on the pod, Ritwik's confirm on -Direct-v0 vs return-to-rest, B3 until an HPC checkpoint exists.
What I am not reporting	No success rate, no "model beats baseline" claim, no training curves. Those wait until scored episodes exist.

What I touch day to day

- Evaluation protocol notes and the fixed-parameter sheet (episode length, Hz, seeds, horizon, versions).
- LightwheelAI/leisaac-pick-orange fields the scorer actually reads.
- Per-orange / per-episode metrics and the results table template.
- Wrapper over LeIsaac policy_inference.py → LeRobot async policy server; one CSV row per episode.
- Failure labels once videos exist: collision, missed grasp, dropped orange, wrong placement, timeout, camera/action mismatch, policy-server disconnect.
- Workbook chapters that sit on this lane (evaluation methodology, implementation progress, schedule).

Week-by-week notes

Hours below are mine only. Dates match how I already tracked the month.

Week 1 | Sep 29 – Oct 5

Focus	Pull every scoring knob out of the workbook so it lives on one sheet
Hours	7
Status	Done

I started with the workbook's evaluation chapter and listed every parameter that has to stay fixed before a score means anything: episode length, step rate, which cameras, the language instruction, how many rounds, how many seeds, action horizon, checkpoint path, simulator version. The workbook calls for 120-second episodes at 30 Hz, at least three pooled runs of five rounds each, seeds recorded and reused. I put that on one sheet so nobody has to dig through the chapter mid-run.

I also wrote that success has to come from the environment's own check, through the LeIsaac-SO101-PickOrange-Direct-v0 variant — not a distance threshold we invent when the built-in check

is inconvenient. I talked to Bharat about the dataset notes and told him which fields I needed from him.

Week 2 | Oct 6 – Oct 12

Focus	Dataset audit + first cut of the metric
Hours	8
Status	Done (one open question left with Ritwik)

Dataset week. I went through LightwheelAI/leisaac-pick-orange and checked what the evaluator depends on: 60 episodes, 30 fps, so101_follower, front and wrist at 480×640, six action/state fields in order shoulder pan, shoulder lift, elbow flex, wrist flex, wrist roll, gripper.

Metric cut: per-orange success = oranges placed ÷ (3 × number of episodes). I also want the per-episode count 0–3 as a histogram — the workbook says the distribution is bimodal, so a mean alone hides the story — plus a 95% interval with numerator and denominator shown so nobody has to guess whether a number came from 3 episodes or 75.

Open problem I have not closed: the demos put three oranges on the plate and then return to rest. If -Direct-v0 only tests placement and ignores the return, our numbers and the published ACT numbers are not the same measurement. I asked Ritwik to check that condition. Still waiting.

Week 3 | Oct 13 – Oct 19

Focus	Evaluation wrapper skeleton + B0/B1 command plans
Hours	9
Status	Skeleton done; no live score yet

I wrote the first version of the evaluation wrapper. It sits on top of LeIsaac's scripts/evaluation/policy_inference.py and talks to LeRobot's async policy server. Inputs: checkpoint ID, policy type, seeds, rounds, episode length, step rate, action horizon, device. Each episode drops one CSV row — run ID, seed, round, checkpoint, horizon, oranges placed, termination reason, simulator version, LeRobot version, video path.

I did not run a scored episode. The single-frame contract test was still pending on Snehil's side and the LeRobot version was not pinned, so any number I printed would have been uninterpretable. Instead I wrote the B0 and B1 run plans so those commands exist and I am not inventing them under time pressure later. That week felt unproductive and was not, mostly.

Week 4 | Oct 20 – Oct 26

Focus	Lock the four result rows, horizon rule, failure labels
Hours	8
Status	Plan locked; labels empty until videos exist

Scoring plan for all four rows. B0 = random action. B1 = lerobot/smolvla_base with no fine-tuning. B2 = external ACT number (44.0% per-orange, 33/75) quoted only — that run was on Isaac Sim 5.1, so we do not pretend we reproduced it. B3 = whatever comes out of the fine-tune, scored with the same script and seeds as B1.

Two decisions I want on the record. First, action horizon gets swept per checkpoint and the chosen value gets published. Default 16 is not sacred; reference work found best values of 70 and 32 on two different checkpoints, and smolvla_base tops out at a chunk of 50. Leaving the default forever quietly burns accuracy. Second, B1 and B3 must share one LeRobot version. The workbook already notes a same-task checkpoint moving from 44% to 27% on a version change alone. If those two rows sit on different versions, I cannot tell training apart from framework.

I also started the failure labeling sheet. Categories: collision, missed grasp, dropped orange, wrong placement, timeout, camera or action mismatch, policy server disconnect. No labels yet — nothing scored.

Week 5 | Oct 27 – Oct 28

Focus	Workbook 2 pack for my chapters + refresh this log
Hours	6
Status	Draft pushed on feature/kanaka_workbook2

Workbook 2 is due. My pieces are Chapter 6 (evaluation methodology), the evaluation half of the system-design writeup, and the bolded updates to implementation progress and the two-semester schedule (Gantt + dependencies). I kept Chapters 3–5 as sync notes only — the group already wrote those in Workbook 1 and they did not need a rewrite. Files are on branch feature/kanaka_workbook2 under workbook/kanaka_wb2/.

Separate from my keyboard, the team's Oct 26 lab log says the contract test now passes: smolvla_base loads at 450.0M and returns six numbers in SO-101 order, chunk (1, 50, 6). Camera map front→camera1 / wrist→camera2 and the dataset instruction string are written down there too. That removes the "do not trust B1" gate I had on Oct 26. It does not give me a B1 score. I still need the A40 rollout and the pod's LeRobot pin before I run anything I would put in a results table.

What I actually own vs what I do not

Own: evaluation protocol notes, run-metadata sheet, dataset audit, metric draft, wrapper skeleton, per-episode CSV schema, B0/B1 run plans, results table template, failure labeling sheet, Workbook 2 evaluation/progress/schedule draft.

Do not own: simulator stack, SmolVLA policy-server path, fine-tune jobs, reference section. This log is supposed to show what I did, not what the team did.

Where things stand (updated Oct 28)

Contract test	Passed on the team side (Oct 26). I can trust a future B1 shape check.
Success condition	Still open — Ritwik to confirm -Direct-v0 vs three-orange + return to rest.
LeRobot version	0.6.1 used on laptop checks. Pod pin not written yet. No scoring until it is.
B0 / B1	Not run. Wrapper + schema + commands ready.
B3	Waiting on HPC checkpoint and a sim-host slot.
Failure labels	Sheet ready; empty until the first videos land.
Workbook 2	My chapters drafted and on the branch for group merge.

Next on my lane

- Get Ritwik's answer on the -Direct-v0 success condition before publishing any rate as final.
- Copy the pod LeRobot version into Lab_log.md; freeze that pin for B1 and B3.
- Day of first rollout: run B0 then B1, commit the CSV under phase1/05_evaluation/.
- Horizon sweep after B1 exists; publish the chosen value with the table.
- Score B3 on the same seeds when the fine-tuned checkpoint arrives.
- Start failure labels from the first saved videos; fold numbers into Workbook 2 before Canvas submit if they land in time.

Repo trail: https://github.com/bharatmanivannan/decentralized-multi-robot-collaboration/tree/feature/kanaka_workbook2

End of log